Values or Errors: optional, variant and expected

How to say "no value" or "it failed" in the type: exceptions versus return values, `optional` for a value that may be missing, `variant` for one of several types and `expected` for a value or an error. Builds on RAII and Exceptions.

Why vocabulary types

A signature like `int find(const char* s)` hides its contract: does `-1` mean "not found"? Is `s` allowed to be null? Who owns the result? A **vocabulary type** puts the answer in the type, so every reader and the compiler see it.

THE QUESTION	THE OLD ANSWER	THE TYPE
Two or three values back from one function?	out-parameters, a throwaway struct	`std::pair`, `std::tuple`, or a named struct
A value that may be absent?	a magic value (`-1`), a null pointer	`std::optional<T>`
One of several different types?	a `union` plus a tag, a base class	`std::variant<A, B>`
A value, or the reason there is none?	an error code plus an out-parameter, an exception	`std::expected<T, E>` (C++23)
A run of elements?	a pointer plus a count	`std::span<T>`, `std::string_view`

All of them are ordinary value types: no heap allocation, no virtual functions, nothing to free.

Exceptions, error codes and optional

SITUATION	USE
a failure the caller cannot sensibly handle here (out of memory, broken invariant, constructor failure)	**exception**: it unwinds to code that can cope
an **expected** failure in normal flow (bad input, "not found", parse error)	return a value that says so: `std::optional`, an error code or enum, `std::expected` (C++23)
a hot path with steady latency (real-time code)	**no exceptions**: error codes or `optional`
a `noexcept` context, or a destructor	never throw: catch inside and log or ignore

```
// failure is expected: no exception
std::optional<int> parseInt(std::string_view s) {
    int v = 0;
    auto r = std::from_chars(s.data(), s.data() + s.size(), v);
    if (r.ec ≠ std::errc{} || r.ptr ≠ s.data() + s.size()) return
std::nullopt;
    return v;
}

// the caller must look at the result
if (auto n = parseInt(input)) use(*n);
else report("not a number");
```

- **Exceptions** keep the normal path clean and cannot be silently ignored; the cost is that a throw is slow and the control flow is invisible.
- **Return values** make failure explicit and cheap; the cost is that every caller must check, and `[[nodiscard]]` helps remind them.
- Constructors cannot return a code: either throw, or use a factory function returning `optional` or an `expected`.
- Be consistent inside a module, and document what a function can throw.

KEY POINT "Exceptions or error codes?" Exceptions for rare failures that need unwinding, return values for expected failures and for hot paths. Mention RAII (so nothing leaks either way) and that a destructor must never throw.

`optional`: a value or nothing

`std::optional<T>` holds either a `T` or nothing. The "nothing" is part of the type, so a caller cannot forget it exists.

```
std::optional<int> parseInt(std::string_view s) {
    int v = 0;
    auto r = std::from_chars(s.data(), s.data() + s.size(), v);
    // "no value"
    if (r.ec ≠ std::errc{} || r.ptr ≠ s.data() + s.size()) return
std::nullopt;
    // converts to optional<int>
    return v;
}

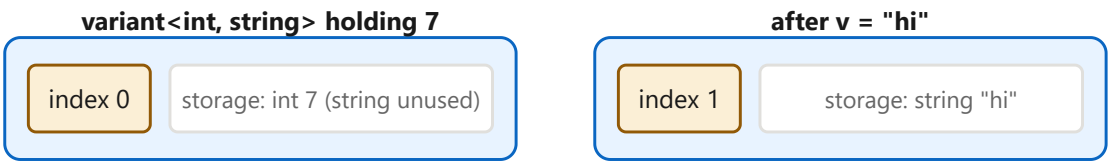
// test, then dereference: n is true when a value is present
if (auto n = parseInt("42")) use(*n);
// -1: a default for the empty case
parseInt("4x").value_or(-1);
parseInt("").has_value();                    // false
// throws std::bad_optional_access
std::optional<int> e;  e.value();
// undefined behaviour: no check, like dereferencing a null pointer
*e;
```

OPERATION	ON AN EMPTY OPTIONAL
`if (o)` / `o.has_value()`	false
`*o`, `o→member`	undefined behaviour: check first
`o.value()`	throws `bad_optional_access`
`o.value_or(x)`	returns `x`
`o.reset()`, `o = std::nullopt`	makes it empty

- The value lives **inside** the optional (compiled: `sizeof(int)` is 4, `sizeof(optional<int>)` is 8: the value plus a flag, padded). No heap allocation, unlike a pointer or `unique_ptr`.
- Use it for a result that is *legitimately* absent: a lookup that may miss, a parse of untrusted input, an optional configuration value. It does not say *why* the value is missing; for that, use `expected`.
- Not a replacement for a reference or pointer that means "refers to an object owned elsewhere": `optional<T&>` does not exist. Use a pointer there.
- Marking a function `[[nodiscard]]` keeps callers from dropping the result.

`variant` and `std::visit`

`std::variant<A, B, C>` is a type-safe union: it holds exactly one of its alternatives and knows which.



Size: the largest alternative plus the index. No heap allocation for the variant itself.

One alternative at a time; the index says which.

```
std::variant<int, std::string> v = 7;
v.index();                                  // 0: the int is active
std::holds_alternative<int>(v);             // true
std::get<int>(v);                           // 7
// throws std::bad_variant_access: the wrong alternative
std::get<std::string>(v);
// a pointer, or nullptr: no exception
if (auto* s = std::get_if<std::string>(&v)) { }
v = std::string("hi");                      // now index 1; the int is gone
// default: holds a value-initialised FIRST alternative (index 0, int 0)
std::variant<int, std::string> d;
```

Visiting every alternative

`std::visit` calls a function with whichever alternative is active. Give it a callable that handles **every** alternative, and a missing one is a **compile error**, not a silent bug.

```cpp
// glue lambdas into one callable
template <class... Ts> struct overloaded : Ts... { using Ts::operator()...; };
// deduction guide (needed before C++20)
template <class... Ts> overloaded(Ts...) -> overloaded<Ts...>;

struct Circle { double r; };
struct Rect   { double w, h; };
// no base class, no virtual functions
using Shape = std::variant<Circle, Rect>;

double area(const Shape& s) {
    return std::visit(overloaded{
        [](const Circle& c) { return 3.0 * c.r * c.r; },
        [](const Rect& r)   { return r.w * r.h; }
    }, s);
}
// a heterogeneous collection stored by value
std::vector<Shape> shapes{Circle{2}, Rect{3, 4}};
// total area: 12 + 12 = 24 (compiled)
```

- **Variant or virtual functions?** A variant fits a *closed* set of types you control (messages, tokens, syntax-tree nodes), where you keep adding operations. Virtual functions fit an *open* set that others may extend.
- If an alternative's constructor throws during assignment, the variant can end up *valueless by exception* (`valueless_by_exception()`). It is rare, and it is why you should keep alternatives cheap and non-throwing to move.
- A reference alternative is ill-formed (use `std::reference_wrapper`). The same type twice is legal, but then you can only reach it by index, not by type.
- `std::monostate` is an empty alternative: put it first to make a variant default-constructible when no alternative should be the default.

`expected`: a value or an error

`std::expected<T, E>` (C++23) holds either the result `T` or an error `E`. It is `optional` that explains *why* the value is missing, and an alternative to exceptions for failures that are normal.

```
enum class ParseError { Empty, NotANumber };

std::expected<int, ParseError> parse(std::string_view s) {
    // the error path
    if (s.empty()) return std::unexpected(ParseError::Empty);
    // the value path
    if (auto n = parseInt(s)) return *n;
    return std::unexpected(ParseError::NotANumber);
}

auto r = parse("12");
if (r) use(*r);                                  // true: it holds a value
else   report(r.error());                        // otherwise: the error
// 0: a default when it holds an error
parse("zz").value_or(0);
// throws std::bad_expected_access<ParseError>
parse("").value();
```

FAILURE STYLE	COST AND VISIBILITY	USE FOR
Exception	invisible in the signature; a throw is slow	rare failures the caller cannot handle here, constructors
Error code plus out-parameter	easy to ignore, clumsy to write	C APIs
`optional<T>`	visible, free, says nothing about why	"not found", where the reason is obvious
`expected<T, E>`	visible, free, carries the reason	expected failures: parsing, I/O, validation

It needs a C++23 standard library (GCC 12 or later, recent MSVC). With an older one, `std::variant<T, E>` or a small result struct does the same job, and the free `tl::expected` library has the same interface. Mark the function `[[nodiscard]]` so the result cannot be ignored.

Which one when

SITUATION	USE
Return two related values that are used right away	`pair` / structured bindings; a named `struct` if they travel further
A lookup that may find nothing	`optional<T>`
A parse or I/O call whose failure has a reason	`expected<T, E>` (or an exception if it is rare)
A value that is one of a few known types	`variant` plus `visit`
A function that reads a run of elements	`span<const T>`; `string_view` for text
A value of any type, unknown at compile time	`std::any` (rare: you lose compile-time checking; prefer a variant)

The same idea runs through all of them: make illegal or missing states **explicit in the type** so the compiler, not a comment, enforces them.

Further reading by section

Where each section of this lesson is explained in more depth. Pages are PDF viewer pages (the number your PDF reader shows).

SECTION	BOOK	PART OF THE BOOK	PDF PAGES
variant and std::visit	C++ Primer	19.5 to 19.6 Nested classes and unions	872 to 880